Software Engineering (SE) Course CSC 648-848 Spring 2026
High Level Description of Class Team Project
Web App: AI LifeOS – Intelligent Student Co-Pilot Platform

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Acknowledge
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Introduction

This document describes at a **very high-level** the software engineering (SE) team **Final Project,** a Web application that student teams will develop this semester. Each student team will design and develop its own version of the same web application which is described in this report. Using only this brief high-level description to initiate the project is on purpose somewhat vague in order to engage students in the full cycle of SW application development starting from scratch and leading to delivery of a final application.

Selection and options for SW development and application deployment tools to be used will be described in the class and in Milestone 0.

Final project organization polices and rules of participation, as well as grading rubrics are described in the class slides about final team project, posted on Canvas. The final team project carries **50% of the class grade**, which each milestone contributes to the final score. In general, all team members get the same grade for the final project (exceptions will be explained in the class, team members who do not participate/collaborate would have less points). Full and maximum focus and participation/attendance to team project are required from each team member.

Student teams are assigned based on academic records and other methods in order to properly mix the skills and experience. Each team will have a formal ***team lead, front and back end engineer, github lead/master and other roles according with their needs and member skills***. Each team will communicate with instructors mainly via their team lead.

Class instructor will serve also as CEOs, CTO and Customer advocates to give feedback on UI. Team leads are encouraged to ask advice from their peers from other teams in the class.

Application to be developed: AI LifeOS – Intelligent Student Co-Pilot Platform

Functional specifications e.g. what functions and services are envisioned –

NOTE: *This is an initial, high-level, vague, and incomplete list meant to get you started. Student teams are expected to refine, improve, and evolve the product vision while preserving the overall purpose and constraints of the system.*

Two entrepreneurs, SFSU alumni, decided to create a start-up comprising a team of about 6+ students to develop a **web-based intelligent platform that acts as an AI-powered co-pilot for college students**, helping them design, manage, and continuously improve their academic, professional, and personal lives.

The web application will function as a **personalized AI life management and discovery system**, integrating **goal setting, intelligent search, reflection, learning analytics, and ethical AI usage guidance** into one unified platform.

The system will allow students to create a **personal dashboard** where they can manage goals, projects, learning activities, and personal development areas such as academic success, career preparation, leadership, wellness, and life skills.

A core component of the platform will be a **powerful search and discovery engine** that enables users to find and explore:

- Goals and project templates
- Productivity and study workflows
- Academic and career resources
- AI tools and learning aids
- Campus-related opportunities and services
- Public community-shared content

Users will be able to **search by keywords and filter** results by multiple criteria such as:

- Category (academic, career, wellness, leadership, finance, creativity, research, etc.)
- Use case (studying, job preparation, mental wellness, skill building, productivity, project management)
- Content type (tools, templates, workflows, reflections, resources, events)
- Skill area (technical, soft skills, leadership, research, health, communication)
- AI-enabled vs. non-AI resources
- Free vs. paid (informational only, no transactions)

Search results may include relevance ranking, tags, ratings, saved items, and side-by-side comparisons where appropriate.
Students will be able to:

- Define short- and long-term goals.
- Organize goals into projects and milestones.
- Attach discovered resources and tools to their goals.
- Track progress visually through dashboards and reports.
- Complete structured reflections (weekly or on-demand).
- Explore categorized collections of resources and AI-supported tools.
- Interact with an optional AI-powered assistant for planning, reflection support, and resource recommendations.

To enhance personalization, users may enter interests, challenges, workloads, and aspirations. The system may use this information and platform activity to:

- Recommend relevant categories, tools, and workflows
- Personalize search results
- Suggest resources aligned with current goals
- Generate insights and growth summaries

The application may include intelligent features such as:

- Goal and project analytics
- Academic workload visualization
- Resource usage tracking
- Reflection trend analysis
- Stress and burnout indicators

- Skills development mapping
- End-of-semester personal growth reports

The system will provide **interactive dashboards**, visual timelines, charts, and personalized reports.
The platform will support community-oriented capabilities, such as:

- Browsing and searching public goal templates and workflows
- Sharing resources, tools, and strategies
- Discovering popular or trending student tools
- Rating, tagging, and bookmarking content
- In-site messaging between users (webmail only)

The platform targets three primary audiences:

- **College Students** – Primary users managing academics, career preparation, and personal development.
- **Faculty & Academic Programs** – Secondary users interested in ethical AI use, learning support, and student development tools.
- **University Support Organizations** – Career centers, tutoring centers, and student success programs that may showcase categorized resources.

General users may explore public content and categorized search features. Registered users will gain access to dashboards, personalization, analytics, saved searches, and collaboration features.
This start-up aims to address the lack of **integrated, intelligent, and ethical AI-supported platforms** that combine **life management with advanced discovery and search capabilities** tailored specifically for college students.

Therefore, student teams are required to design **unique and innovative search-related features** such as smart filters, AI-assisted search, personalized discovery feeds, ethical AI labeling, intelligent tagging, or visual resource maps.

The system must be designed with **modularity, maintainability, and scalability** in mind, particularly around the **search, data modeling, and analytics components**, to allow future expansion into new datasets, AI services, and institutional partnerships.

This WWW service must be very easy to use and must be created in such a way as to facilitate application usage (e.g., media-rich dashboards, interactive search interfaces, dynamic filters, visual exploration tools, maps, and intelligent recommendations).

For this class, the application will focus on **desktop/laptop browsers**, and NOT on native mobile application development. However, the system must use **responsive UI implementation** so that it renders well on mobile devices.

The users of the system are assumed to have adequate Internet connectivity.

Important constraints

- Due to site security issues and limitations of cloud servers where the application will be developed and deployed, **no external mail clients shall be used**. Messaging between users must be done via **in-site messaging (webmail)** only.
- **No cost or payment functionality** will be implemented or simulated in the user interface.
- To verify a proper email address from users, the application **may require email confirmation** through system-generated messages.

High-level non-functional specifications (how the app is delivered and other constraints) that MUST be adhered to

(How the app is delivered and other constraints that MUST be adhered to)

1. Application shall be developed, tested, and deployed using tools and servers approved by Class CTO and as agreed in M0.
2. Application shall be optimized for standard desktop/laptop browsers and must render correctly on the two latest versions of at least two major browsers.
3. Selected application functions must render well on mobile devices (this is a plus).
4. Data shall be stored in the team's chosen database technology on the team's deployment server.
5. Privacy of users shall be protected, and all privacy policies shall be clearly communicated to users.
6. The language used shall be English.
7. Application shall be very easy to use and intuitive.
8. Google Maps and Google Analytics shall be added where appropriate (e.g., campus resources, study locations, event discovery, usage analytics).
9. No external e-mail clients shall be allowed. You shall use in-site webmail only.
10. Pay functionality, if any, shall not be implemented nor simulated in the UI.
11. Site security: basic best practices shall be applied (as covered in the class).
12. Modern Software Engineering processes and practices shall be used as specified in the class, including collaborative and continuous software development.
13. The website shall prominently display the following exact text on all pages at the top of the WWW page:
    "SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only"
14. AI-powered assistant and intelligent search enhancements are **optional (bonus requirements)** and must be implemented modularly.

These non-functional specs are in real life usually provided by clients, CEO, business and legal departments and are NOT subject to change by engineering team on their own. Hence copy them in your milestone documents (and optionally add details as necessary). **You are not allowed to remove any of these requirements by yourself and must abide by them exactly as they are written (including the text for # 14).**

Data and Content

The website has to be media rich in order to facilitate adoption of this service. However, since the projects will be posted for outside viewers, students must ensure that they have legal rights to use all posted content (including logos and background imagery). Students are encouraged to make up their own content and data.

Images for posting shall be made and owned by students who develop the application or be from a royalty free source. Item descriptions and prices can be made up by students.

The content (e.g., images, text) itself has to abide by ethical principles and guidelines for on-line communications at SFSU. Teams must ensure that no inappropriate material (text or images) is presented on their site.

Enough mapping and images should be provided for the demo such that you can effectively demo search and filtering functions. Please plan accordingly for display by the corresponding decided categories for a successful demo.

Have fun, ask lots of questions, and produce a great web app! And, as you will learn in the class, always think of users first!